DESIGN AND FABRICATION OF MECHANICAL APPARATUS FOR 3D RECONSTRUCTION

A PROJECT REPORT

Submitted by

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A note on this web edition (2026)

This is the B.E. project report submitted to Anna University in November 2024, edited in 2026 for publication on my website. Everything not listed below is unchanged from the submitted version.

1. Student register numbers have been removed.

2. Eight entries have been removed from the literature survey in Chapter 2: the 2023 papers attributed to Zhang et al., Kim et al., Martinez et al., Singh et al., Wilson et al., Chen et al., Brown et al. and Lee et al. I could not trace them to real publications, and none of them appears in the reference list.

3. Table 7.1 originally said the reconstruction methods were tested under similar conditions. Its figures are values published by each method's authors on their own benchmarks, not our measurements, and some do not match their sources. The table has been relabelled and should not be cited.

4. Several volume and page numbers in the reference list do not match the original publications. Please cite the original papers rather than this list.

5. Some explanatory figures in the background chapters are reproduced from the published papers that describe those methods, and are not our own work.

The measurements that are our own are unchanged: the image registration trials comparing handheld capture with the rig, and the Instant-NGP timings, both in Chapter 7.

Anush Bharathi R

BONAFIDE CERTIFICATE

Certified that this project report **"DESIGN AND FABRICATION OF MECHANICAL APPARATUS FOR 3D RECONSTRUCTION"** is the bonafide work of **"ANUSH BHARATHI R, KAMRAN B and PATHMARAJAN R"** who carried out the project work under my supervision.

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ABSTRACT

Three-dimensional (3D) reconstruction has garnered significant research interest over the past decade due to its growing demand in fields like reverse engineering, robotics, and virtual reality. Conventional techniques such as Shape from Motion (SFM), Shape from Shading (SFS), stereo vision, and structure from silhouettes primarily calculate depth by identifying and comparing corresponding points between pixels across multiple frames. However, these methods are computationally expensive, require large datasets, and involve prolonged processing times, making them unsuitable for real-time or resource-constrained applications.

In this project, a novel mechanical fixture has been designed and developed to address these challenges. The fixture, inspired by a serial manipulator, consists of two joints and two links, with a camera holder as its end effector. The setup rotates around a centrally placed object, and the camera's position is precisely calculated using Denavit-Hartenberg (DH) parameters, supported by joint angle sensors for tracking motion in the X, Y, and Z axes. The data collected from these sensors, along with captured images, serves as input to advanced AI-based 3D reconstruction models.

This project evaluates state-of-the-art techniques such as Neural Radiance Fields (NeRF), Instant Neural Graphics Primitives (INGP), and Gaussian Splatting, which offer faster processing times, real-time rendering capabilities, and comparable or superior reconstruction quality compared to traditional approaches. By integrating the mechanical fixture with these AI models, the project aims to significantly reduce computational requirements and reconstruction times while maintaining high accuracy and quality in 3D representation.

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CHAPTER 1

INTRODUCTION

1.1 THE ORIGINS OF 3D RECONSTRUCTION

3D reconstruction is the process of creating digital 3D representations of physical objects or environments using input data like 2D images, point clouds, or depth measurements. This field has evolved over centuries, beginning with ancient artistic and geometric practices and advancing into sophisticated computational methods.

Early forms of 3D modeling can be seen in ancient civilizations. The Egyptians demonstrated an understanding of spatial form through detailed statues and bas-reliefs in structures like the Pyramids and temples at Karnak and Luxor. Similarly, ancient Greek thinkers like Euclid and Pythagoras laid the groundwork for spatial reasoning with their pioneering work in geometry.

The Renaissance marked a major leap in 3D visualization techniques. Filippo Brunelleschi introduced linear perspective in 1415, enabling realistic depth representation on flat surfaces. This was later formalized by Leon Battista Alberti in his 1435 book *Della Pittura*, providing practical guidelines for depicting spatial depth that laid critical groundwork for 3D reconstruction. The invention of photogrammetry by Aimé Laussedat in the 1850s advanced 3D reconstruction. Using overlapping photographs, this technique enabled accurate 3D mapping for cartography, architecture, and engineering, establishing a foundation for modern methods.

The advent of computers revolutionized 3D reconstruction. In the 1960s, MIT's Roberts developed methods for extracting 3D structures like cubes and prisms from digital images. Marr and Poggio further expanded this in the 1970s, proposing frameworks for reconstructing 3D structures from 2D images by addressing challenges like stereo matching and parallax.

In the 1980s and 1990s, structured light projection and laser scanning emerged as highly accurate methods for 3D measurement. Structured light uses projected patterns to map surfaces, while laser scanning captures detailed data for applications in industrial inspection and robotics.

1.2. EXPRESSION OF 3D RECONSTRUCTION

3D reconstruction is divided into explicit and implicit expression methods based on different approaches, offering diverse perspectives and processing techniques for data obtained from the real world.

1.2.1 Explicit Expression

The main methods for displaying 3D data are point clouds, voxels, and meshes. Point clouds are unordered collections of discrete data points in 3D space, often used to represent the external surface of objects or spatial scenes. Voxels divide 3D space into uniform cubic units, each containing spatial attributes like color or density, and are commonly used in medical imaging and fluid dynamics, though they require significant storage. Meshes, composed of vertices, edges, and faces (often triangles), accurately represent complex geometric shapes and details, making them ideal for computer graphics and engineering due to their efficiency in rasterization and suitability for representing planar surfaces.

1.2.2 Implicit Expression

Implicit expressions represent 3D models without explicitly storing geometric data, saving storage space and enabling complex shape processing. While precise and adaptable, implicit representations can be challenging to compute, understand, and manipulate. Common types include implicit surfaces (e.g., Bézier curves, NURBS), Signed Distance Functions (SDF), occupancy fields, and radiance fields. SDFs use signed distances to represent object geometry, with values indicating a point's position relative to the surface

(inside, outside, or on it). Occupancy fields map spatial points to attributes, while radiance fields link points and rays to density and color. Implicit methods allow shape recovery at any resolution and are highly flexible for modeling complex geometries. Combining implicit and explicit methods leverages their respective strengths—flexibility and storage efficiency of implicit methods with the intuitiveness and precision of explicit representations—optimizing 3D modeling and analysis for various applications.

1.3 EXISTING TRADITIONAL 3D RECONSTRUCTION METHODS

3D reconstruction methods are classified into contact and non-contact data acquisition techniques. Contact methods involve instruments like coordinate measuring machines (CMMs) that physically measure the 3D scene through trigger or continuous measurements, suitable only for objects that can be directly contacted. Non-contact methods, however, gather data without physical contact by using visual sensors to capture image sequences. These methods are further divided into active and passive techniques. Active methods interfere with the object using technologies like structured light, laser scanning, and time-of-flight (TOF) to create depth maps. Passive methods rely solely on optical sensors to measure radiance and infer 3D structures without interacting with the object.

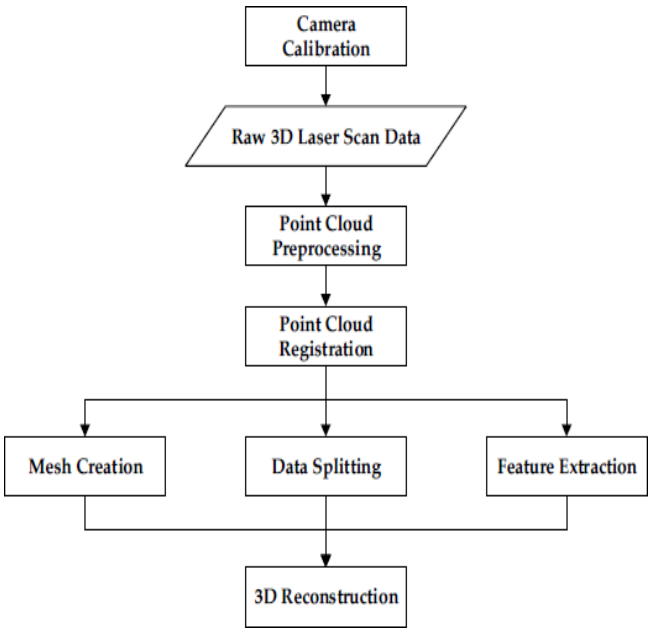
1.3.1 Laser Scanning

Laser scanning uses a laser beam combined with a rangefinder to measure reflections or scattering, calculating distances on the target's surface. It quickly captures detailed surface shapes as point clouds, enabling accurate 3D modeling and visualization with high precision and density. However, it struggles with reflective or transparent surfaces. Triangulation laser scanning uses a setup where the laser point, camera, and emitter form a triangle. Laser scanning is widely used in terrain mapping, architecture, cultural heritage

preservation, manufacturing, and virtual reality, and is integral to Building Information Modelling (BIM), offering 3D records for engineering and

maintenance. It is also employed in robotics, holography, and 3D holographic projection for immersive visuals. The Figure 1.1 represents the Block Diagram for Laser Scanning.

Figure 1.1 Block Diagram for Laser Scanning



1.3.2 Structured Light

The structured light method projects a patterned light onto a scene and captures its deformation with a camera to infer the 3D structure of an object's surface. Widely used for precise surface mapping, the structured light triangulation principle was enhanced with a grating structure method to address occlusion issues. The Figure 1.2 represents Structured light Triangulation.

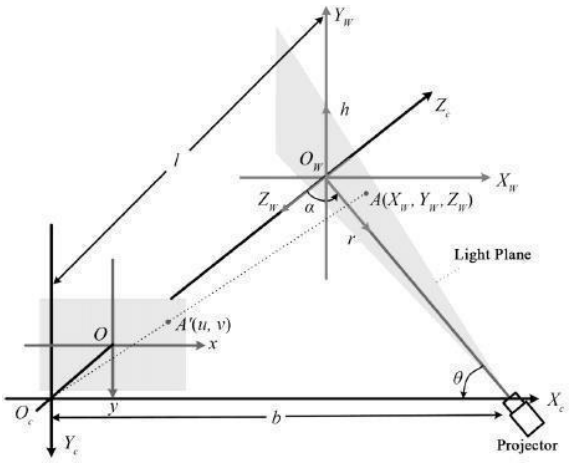


Figure 1.2 Structured light Triangulation

Improved quality of light stripe images was achieved by controlling exposure time, fusing multiple images, and optimizing exposure based on surface analysis. Dynamic range was further enhanced using Liquid Crystal on Silicon (LCoS) technology. A cubic polynomial curve fitted the centerline of the light stripe, improving center extraction accuracy. While the method offers strong real-time performance, it is less effective in environments with extreme lighting conditions.

1.3.3 Binocular Stereo Vision

The stereo vision method uses binocular cameras to capture different angles of a scene, calculating depth by analyzing parallax between matching feature points in the images. As depth increases, parallax decreases, meaning larger depth ranges correspond to smaller parallax values. Binocular vision is cost-effective and suited for short-range measurements, though it requires high texture quality. It is widely used in robotics for extracting 3D object positions near autonomous systems. The Figure 1.3 represents Block Diagram for Binocular Stereo Vision.

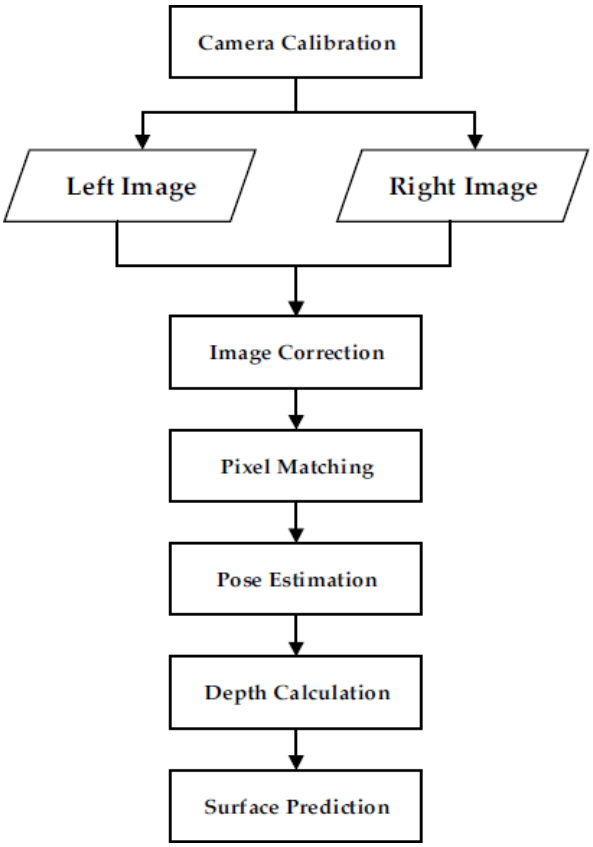


Figure 1.3 Block Diagram for Binocular Stereo Vision

In the binocular stereo vision system, epipolar geometry describes the relationship between the optical centers of the left and right cameras connected by a baseline B, which serves as the rotation axis of the epipolar plane cluster.

The intersection of the epipolar plane with the left and right image planes forms the left and right epipolar lines (l0 and l1). The projection points (x0 and x1) of the observed object onto the left and right image planes lie on these lines, reducing the search range for matching points and simplifying depth computation. The Figure 1.4 represents Epipolar Geometry Recognition.

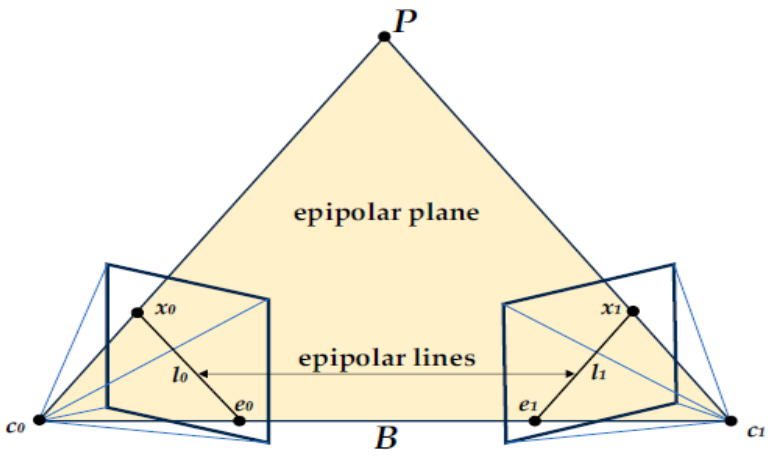


Figure 1.4 Epipolar Geometry Recognition

1.3.4. Structure from Motion (SFM)

This method utilizes the relationship between camera motion and scene structure to reconstruct 3D scenes from image sequences captured at different times or locations. Structure from Motion (SFM) uses overlapping photos of the same object taken from various perspectives as input, outputting a 3D reconstruction along with the camera's internal and external parameters. The Figure 1.5 represents Multi-view Stereo Images.

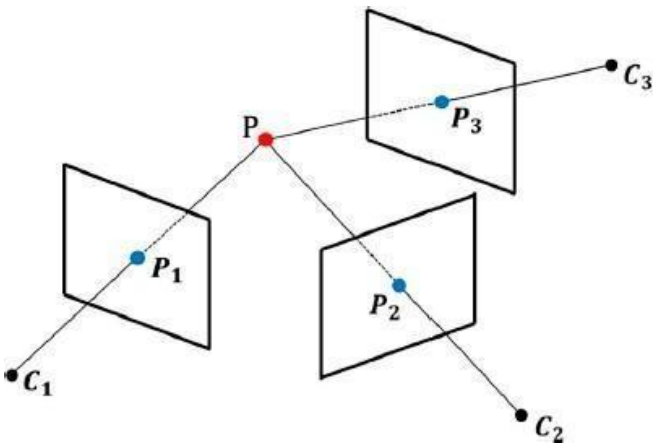


Figure 1.5 Multi-view Stereo Images

Multi-view geometry reconstructs 3D structures by observing scenes from multiple perspectives, using high-quality images with sufficient overlap. It addresses challenges like viewing angle variation, shadows, and image registration to maintain consistent geometric information. Camera calibration is critical for achieving accurate and reliable 3D reconstruction.

1.4. DYNAMIC 3D RECONSTRUCTION METHODS

Dynamic 3D reconstruction captures and models the 3D structure of objects and environments in dynamic scenes, handling factors like object motion, lighting changes, and scene evolution. It uses techniques such as feature point matching to track key features and motion estimation to determine camera motion across frames.

Multi-View Dynamic 3D Reconstruction

This method uses multiple cameras to observe a scene from various perspectives, capturing simultaneous images with consistent timestamps. Feature points or descriptors are extracted and matched to establish correspondences between images. Using pose information and scene structure, 3D models or point clouds are reconstructed and optimized through motion estimation and noise removal. This approach is commonly used to analyze human posture, providing data like joint angles and body proportions.

Simultaneous Localization and Mapping (SLAM)

SLAM builds or updates maps of unfamiliar environments while tracking targets' locations using visual and inertial sensors for improved motion estimation. In dynamic environments, SLAM addresses challenges like loop closure detection to correct accumulated mapping errors. It can run in real time (online SLAM) or process collected data later (offline SLAM), iteratively updating maps for accurate navigation.

1.5. EXISTING MECHANICAL SETUPS

The basic vision system includes a WATEC CCD camera, image processing software, an image processing board, and a computer. The CCD camera captures surface images at 768×574 resolution with 1/30 s grabbing speed and 8-bit output. The experimental rig comprises a 1m×0.5m rigid base supporting an adjustable specimen mount table, a diffuse light source, and a vertically mounted CCD camera focused on a circular table. The light source's mount allows adjustments for angle and distance, and the table can rotate to align striations at specific angles. However, the setup limits specimen size due to the camera's fixed position in the curved frame and requires repositioning the camera to capture images from all sides. The Figure 1.6 represents Diagram of Vision System Setup.

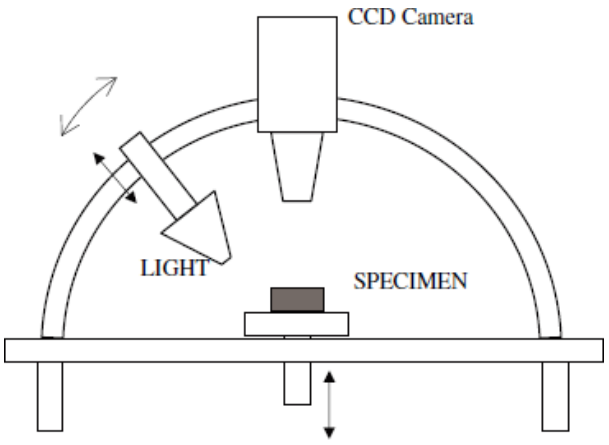


Figure 1.6 Diagram of Vision System Setup

The proposed system enables automated 3D model digitization and precise multi-sensorial scanning, consisting of a 4-DoF mechanical arm, a rotary table, and various scanning sensors. The setup includes a rotary table for holding the object and a depth sensor (Intel RealSense SR300) mounted on the arm's end-effector. Controlled by a Windows OS PC, the system

rotates the object in specific increments (e.g., 5°) for a complete 360° scan. While efficient, the system is costly due to its automated components and depth sensor. Additionally, the mechanical arm lacks the sophistication to capture intricate shapes and curved surfaces effectively. The Figure 1.7 represents Multi sensorial Scanning.



Figure 1.8 Multi-sensorial Scanning

1.6. CHALLENGES IN EXISTING METHODS

Traditional 3D reconstruction methods have several limitations, including sensitivity to environmental factors and the complexity of data acquisition. Techniques like laser scanning and structured light are ineffective for reflective or transparent materials, while photogrammetry depends on high-quality, overlapping images, which can lead to inaccuracies in large or complex scenes. Methods like multi-view stereo require significant computational power to process dense point clouds and manage occlusions, limiting their efficiency and applicability in various environments. These challenges hinder the accuracy and versatility of traditional 3D reconstruction approaches.

1.7. OUR APPROACH

This project addresses these limitations by developing an innovative mechanical apparatus integrated with state-of-the-art (SOTA) AI models for efficient 3D reconstruction. The mechanical system consists of a sophisticated multi-arm setup with four degrees of freedom.

The approach involves design and fabricating a real-time camera position And Angle Tracking System That Can Produce Direction Vectors for Each Image frame captured by the camera. The apparatus involves integration of rotary encoders which help in determining the change in angular positions of the linkage arms.

On the software side, the project explores AI-based reconstruction approaches, including Neural Radiance Fields (NeRF) and Instant Neural Graphics Primitives (INGP). These modern AI techniques offer significant advantages over traditional methods such as

- Faster processing times
- Comparable or superior reconstruction quality
- Novel approaches to 3D representation
- Real time rendering capabilities

The integration of the mechanical tracking system with these AI models aims to achieve two primary objectives:

- Significant reduction in reconstruction time compared to traditional photogrammetry methods.
- Maintenance of high-quality reconstruction accuracy through precise camera position tracking.

CHAPTER 2

LITERATURE SURVEY

This chapter presents a comprehensive literature survey on vision-based 3D reconstruction methods, robotic systems, and actuator mechanisms, emphasizing advancements, methodologies, and their potential applications across various fields. It highlights the significance of integrating traditional and AI-driven techniques, low-cost educational systems, and innovative actuator designs to overcome challenges in dynamic environments and improve computational efficiency. This survey provides a detailed understanding of the current state of the art and establishes a foundation for further advancements and broader applications. The following research papers have been studied to fulfill our objectives.

2.1 REVIEWED PAPERS

Linglong Zhou et al. (2024), Beijing Information Science and Technology University presented a paper titled “A Comprehensive Review of Vision-Based 3D Reconstruction Methods.” This review explores the advancements in 3D reconstruction technologies, categorizing them into traditional static, dynamic, and machine-learning-based approaches. It highlights applications in fields such as robotics, virtual reality, and medicine, and discusses challenges like handling dynamic changes, optimizing accuracy, and enhancing computational efficiency. The study concludes by summarizing the prospects and obstacles for 3D reconstruction technologies.

Haitao Luo et al (2024), presents a detailed review paper of large-scale 3D reconstruction techniques from multi-view imagery, focusing on remote sensing and computer vision applications. The paper organizes 3D reconstruction methods into traditional and learning-based categories, including both active

techniques like structured light and laser scanning, and passive methods like Structure from Motion (SfM), stereo matching, and multi-view stereo (MVS). The review also introduces novel neural-radiance-field-based 3D reconstruction approaches, emphasizing the workflow improvements specific to large-scale scenes. A comprehensive discussion on datasets and evaluation metrics for outdoor and large-scale 3D reconstruction follows, highlighting key challenges like handling lighting, dynamic objects, and computational demands in vast scenes. This paper provides insights into advancements in both traditional and AI-driven techniques, pointing to future directions such as improved neural-network-based methods for more efficient and accurate large-scale reconstructions in fields like urban planning, navigation, and cultural heritage preservation.

Kyriaki A. Tychola et al (2022), International Hellenic University, Greece proposed a review paper on "3D Reconstruction Using RGB-D Cameras". The paper discusses the working of RGB-D cameras, which combine color imaging (RGB) with depth sensing (D) to create detailed 3D models of environments and objects in real-time. It explains the functioning of these cameras, which project infrared patterns to measure depth. While RGB-D cameras offer benefits like low cost and effective indoor mapping, they face limitations including performance challenges in outdoor settings, accuracy issues with reflective surfaces, and high processing requirements.

Julián Tachella et al. (2019), Heriot-Watt University presents a paper on novel "Real-time framework for 3D scene reconstruction using single-photon lidar data". This method addresses a significant limitation in depth imaging through challenging environments by enabling fast 3D reconstruction with plug-and- play point cloud denoisers. Leveraging statistical models and scalable computational techniques, the system achieves 3D reconstructions of complex

outdoor scenes within 20 ms, from distances up to 320 meters, even in daylight. The framework adapts point cloud denoising algorithms to lidar data, efficiently managing multiple surfaces per pixel for high-speed, accurate target reconstruction in dynamic environments. This advancement has applications in fields requiring real-time 3D imaging, such as autonomous navigation and environmental monitoring, and demonstrates a substantial improvement in processing time and robustness over existing methods.

Todd Macdonald and Wilbur Yang (2014), Stanford University presents a paper on "3D Image Reconstruction from Stereo Images and Single Images". This paper discusses a method for 3D scene reconstruction using stereo images, aiming to create semantically plausible 3D scenes without active depth sensors. Their approach involves segmenting objects from stereo images, calculating depth from disparity maps, and matching these segmented shapes with corresponding 3D models from a pre-built database using Zernike descriptors for rotation and scale invariance. The reconstructed 3D scenes integrate these selected models, preserving orientation and depth. This model-based approach facilitates realistic 3D reconstructions for applications like virtual reality and potentially extends to single-image reconstructions if paired with a labeled dataset for real-world scaling. The study explores improvements over purely point-based reconstructions by using 3D models, enhancing both visual coherence and computational efficiency. Future directions include refining segmentation accuracy and addressing occlusion challenges, potentially with semantic graphs.

Tanzila Younas et al (2019) SZABIST, Karachi, Pakistan, proposed a paper on "Four Degrees of Freedom Robotic Arm". This paper addresses the fabrication of a cost effective robotic system with 4-revolute joints for educational purposes. The paper focuses on the implementation of Forward Kinematics

algorithms to compare calculated and real-time positions of the arm, enabling better understanding of these concepts. The paper summarises the system design and kinematic modelling using DH convention.

Amin A. Mohammed and M. Sunar (2015), King Fahd University of Petroleum & Minerals, Dhahran, Saudi Arabia presented a paper on "Kinematics Modeling of a 4-DOF Robotic Arm".This paper highlights the usage of two methods for the kinematic modelling of a robotic arm namely DH method and product of exponentials formula. It also addresses the usage of inverse kinematics by finding the angular positions through position equations.

Jae Sik Lim and Young Jin Lee (2014), Korea Electrotechnology Research Institute and Autopower Co., Ltd, presented a paper on "Detection of Absolute Position of Robot Actuator with Two Incremental Encoders". This paper proposes an efficient method for detecting absolute robot arm positions using dual incremental encoders considering a robot joint comprising a motor, a reducer, two encoder, and motor driver. The key innovation lies in using two encoders, one at the motor input (first encoder) and another at the reducer output (second encoder) to create a unique relationship table between their count values. The paper presents two approaches: one using relative prime numbers between reducer ratio and encoder pulses, and another using irregular encoder disks.

2.2 NEED FOR STUDY

Three-dimensional reconstruction has seen remarkable advancements through various methodologies, ranging from traditional photogrammetry to modern neural approaches. Recent comprehensive reviews by Zhou et al. (2024) highlight that while neural methods like Instant-NGP (Müller, 2022) have reduced processing times from days to minutes, they still heavily depend on Structure-from-Motion (SfM) for camera pose estimation, creating a significant bottleneck in the reconstruction pipeline. The integration of RGB-D cameras, as studied by Tychola et al. (2022), offers real-time capabilities but faces limitations in outdoor settings and with reflective surfaces. While automated mechanical scanning

solutions exist, as demonstrated by Rousopoulou (2019), they often require expensive hardware setups or sacrifice precision for automation. Research in robotic arm kinematics by Younas et al. (2019) and Mohammed and Sunar (2015) demonstrates the potential of mechanical systems for precise position tracking, while Lim and Lee (2014) shows the effectiveness of dual encoder systems for accurate position detection However, no existing solution successfully combines mechanical precision with state-of-the-art neural reconstruction methods in a practical, affordable system. There is therefore a clear need for an integrated approach that leverages both mechanical tracking precision and efficient neural reconstruction capabilities to achieve faster, more accurate 3D reconstruction while maintaining affordability and accessibility.

CHAPTER 3

OBJECTIVE AND METHODOLOGY

3.1 OBJECTIVE

To design and fabricate a 4-DOF a real-time mechanical camera position and angle tracking system that can produce direction vectors for each image frame captured by the camera**.** The apparatus prioritizes reducing processing time, improving accuracy by extracting angular coordinates and functioning under diverse lighting conditions.

3.2 METHODOLOGY

Methodology refers to the specific steps, tools and procedures used to collect and analyse the data to obtain the results. The process involved in the completion of the design and fabrication project is shown in Figure 3.1.

The methodology involves the following steps:

- System Architecture Design and Analysis.
- Position Tracking System Development
- Mechanical System Fabrication
- Electronic System Integration
- Software Framework Development
- Neural Reconstruction Pipeline Integration
- Calibration and Error Compensation
- Performance Evaluation and Optimization

CHAPTER 4

DESIGN AND FABRICATION OF PROTOTYPE

4.1 DESIGN OVERVIEW

The project's motive is to design and fabricate an innovative mechanical apparatus integrated with state-of-the-art (SOTA) AI models for efficient 3D reconstruction. Such an apparatus would be scalable according to the target size requirements. For the sake of prototyping, we have arbitrarily chosen a $30cm^3$ of object volume. To ensure multiple viewpoints to be captured, the work volume has to be more than $90cm^3$. The mechanical system consists of a sophisticated multi-arm setup with four degrees of freedom that has a ,maximum radius of 136cm(hence, ensuring a $90cm^3$ work volume). The figure 4.1 depicts the overview image of the apparatus which has been designed.

Figure 4.1 Overview of 4-DOF camera tracker design

4.2 DESIGN CALCULATIONS

The dimensions of the linkage system were determined based on the required workspace volume and operational constraints:

1. Workspace Volume Determination

a. Target object volume: 30 cm^3
b. Required minimum workspace: 90 cm^3 (to ensure complete coverage)
c. Minimum viewing distance: 40 cm (for optimal focus and field of view)

2. Linkage Size Calculation

The length of each arm (L = 32 cm) was determined using:

a. Required maximum reach (R): 136 cm
b. Base offset (B): 40 cm
c. Number of linkages (n): 3
d. Individual link length: $L = (R - B)/n = (136 - 40)/3 \approx 32$ cm

3. Joint Plate Dimensions

a. Base joint plate width (W1): 8.4 cm
    i. Determined by encoder mounting requirements (4.2 cm per side)
b. End joint plate width (W2): 4.7 cm
    i. Optimized for stability while minimizing weight

4. Height Offset Calculation

a. Base height (H): 31.5 cm
    i. Determined by: $H = \sqrt{(R^2/4 + B^2)}$
    ii. Ensures optimal viewing angles between 30° and 60°
    iii. Provides clearance for full rotation

During position tracking, for each quadrilateral linkage, the following parameters are needed to be determined in real time:

- The effective length of the equivalent single arm
- The angle of this equivalent arm relative to the base joint plate

Let's define our variables:

- $\theta 1$, $\theta 2$: The two angles measured from the base joint plate to the sides of the quadrilateral

- L1, L2: The lengths of the two sides of the quadrilateral

- W1, W2: The widths of the joint plates at the start and end of the linkage

- Le: The effective length of the equivalent single arm
- θe: The effective angle of the equivalent single arm Steps involved in

formation of kinematic equations:

Step 1: Calculate the coordinates of the end points

Let's set up a coordinate system with the origin at the midpoint of the base joint plate:

- Start point: (0, 0)
- End point of first arm: (L1 * cos(θ1), L1 * sin(θ1))
- End point of second arm: (L2 * cos(θ2), L2 * sin(θ2)) Step 2: Calculate

the midpoint of the end joint plate

The midpoint (x, y) of the end joint plate is:

- x = (L1 * cos(θ1) + L2 * cos(θ2)) / 2
  (4.1)
- y = (L1 * cos(θ1) + L2 * cos(θ2)) / 2
  (4.2)

Step 3: Calculate the effective length (Le)

The effective length is the distance between the midpoints of the start and end joint plates:

- Le = sqrt(x^2 + y^2)
  (4.3)

Step 4: Calculate the effective angle (θe)

The effective angle is the arctangent of the y and x coordinates of the end midpoint:

- θe = atan2(y, x)

(4.4)

4.2.1 Kinematic Equations:

These equations define the position of the end-effector of the multi-jointed system, in terms of various joint angles and link lengths.

$$x = 40*\cos(\theta b) + Le1*\cos(\theta b)*\cos(\theta e1) + Le2*\cos(\theta b)*\cos(\theta e1+\theta e2) + Le3*\cos(\theta b)*\cos(\theta e1+\theta e2+\theta e3) \quad (4.5)$$

$$y = 40*\sin(\theta b) + Le1*\sin(\theta b)*\cos(\theta e1) + Le2*\sin(\theta b)*\cos(\theta e1+\theta e2) + Le3*\sin(\theta b)*\cos(\theta e1+\theta e2+\theta e3) \quad (4.6)$$

$$z = 31.5 + Le1*\sin(\theta e1) + Le2*\sin(\theta e1+\theta e2) + Le3*\sin(\theta e1+\theta e2+\theta e3) \quad (4.7)$$

4.2.2 Effective Length (Lei):

This equation defines the **effective length** of a system of links. The idea here is that multiple joint angles affect how "effective" a link's length is when projecting in the x, y, or z directions.

$$Lei = 32 * \text{sqrt}((\cos(\theta i1) + \cos(\theta i2))^2 + (\sin(\theta i1) + \sin(\theta i2))^2) / 2 \quad (4.8)$$

4.2.3 Effective Angle (θei):

This equation defines the **effective angle** of the system, based on the joint angles. The tan function computes the angle from the horizontal, taking into account both the x and y components of the system:

- The sine terms affect the y-component, while the cosine terms affect the x-component.
- The result gives the overall angle after combining the individual joint angles' effects on the system's position.

$$\theta ei = \text{atan2}(32*(\sin(\theta i1) + \sin(\theta i2)), 32*(\cos(\theta i1) + \cos(\theta i2))) \quad (4.9)$$

The usage of the above equations help in determining the linear and angular positions of the end effector

4.3 COMPONENTS

The components required for the implementation of project includes:

1. Base
2. Bearing Casing
3. Clamps
4. Shaft
5. Circular Platform
6. Roller Support
7. Joint-1
8. Arm-1
9. Joint-2
10. Arm-2
11. Joint-3
12. Arm-3
13. Joint-4
14. Camera Holder

4.3.1 Base

The base acts as a support platform for the entire structure. This Figure 4.2 depicts the image of the base.

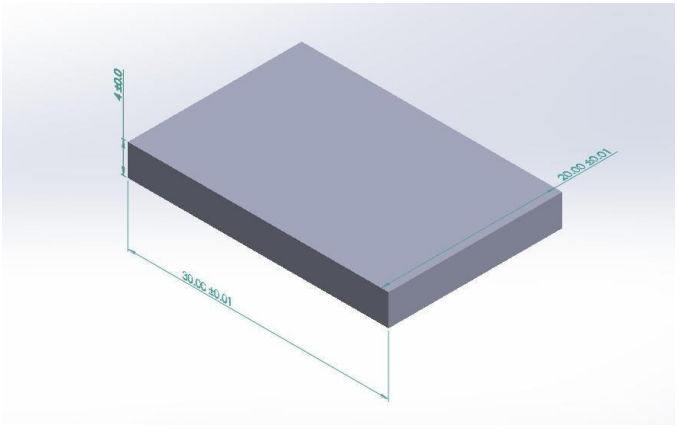


Figure 4.2 Base

4.3.2 Bearing Casing

The bearing casing acts as an enclosure to the central bearing. The bottom link is attached to the bearing casing. It rotates along with the casing. This Figure 4.3 depicts the image of the bearing casing.

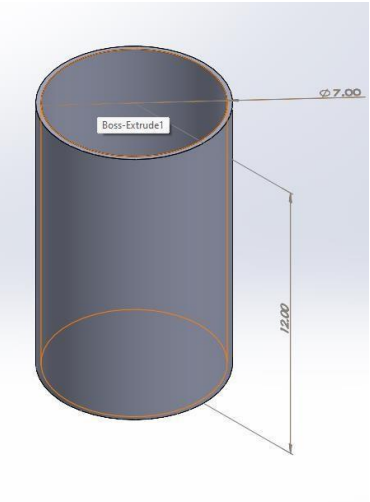


Figure 4.3 Bearing Casing

4.3.3 Clamps

The clamps are attached at various positions to hold the links together with the bearing casing and to hold the rotary encoders. This Figure 4.4 depicts the image of the clamps.

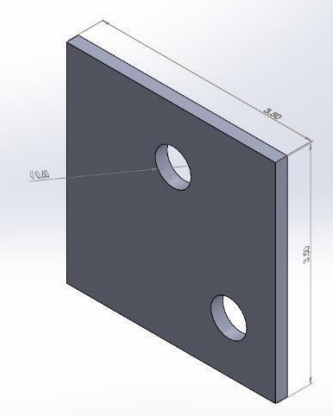


Figure 4.4 Clamps

4.3.4 Shaft And Circular Platform

The shaft is used to join the circular platform and the base platform. The shaft is inserted in the bearing and is fixed. This Figure 4.5 depicts the image of the shaft. The circular platform is used to place the object that is needed to be reconstructed. This Figure 4.6 depicts the image of the circular platform.

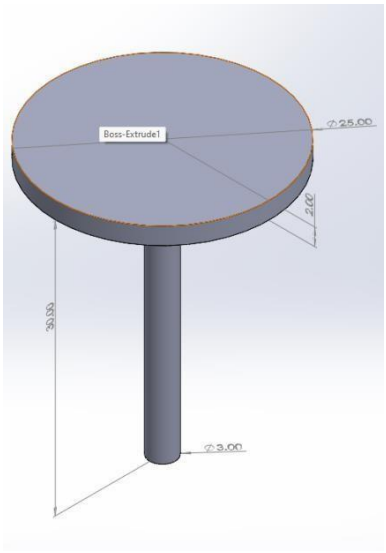


Figure 4.5 Shaft and Circular platform

4.3.5 Roller Support

The roller support consists of four wheels and it is held to a rectangular block. It provides better stability to the structure. This Figure 4.7 depicts the image of the roller support.

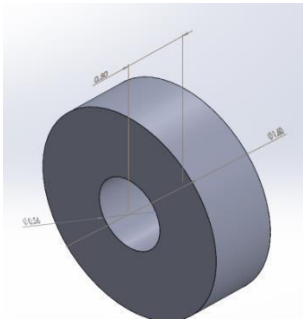


Figure 4.6 Roller Support

4.3.6 Joints

The joints are triangular and penta-lateral in shape and are used to hold various links together. This Figure 4.8 depicts the image of a joint.

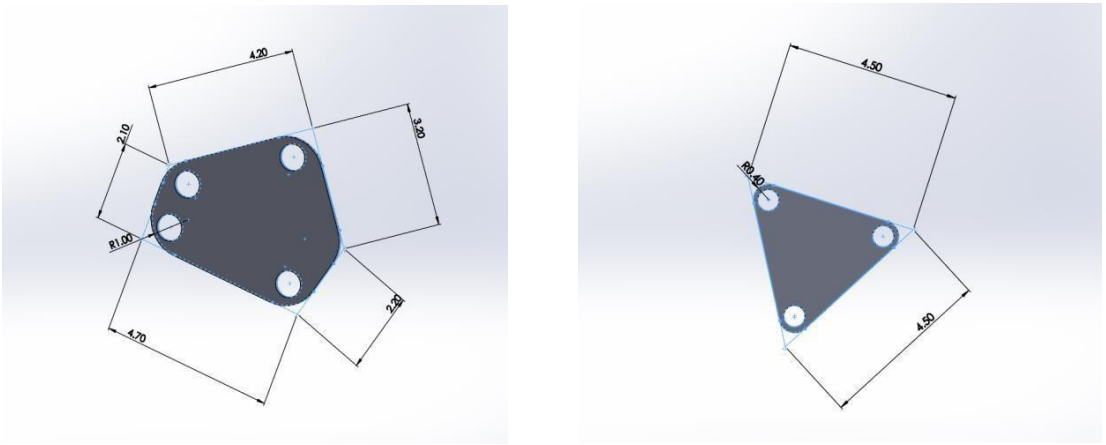


Figure 4.7 Joint

4.3.7 Camera Holder

The camera holder is used to hold the camera which is used to take the video of the entire object placed on the circular platform. This Figure 4.8 depicts the image of the camera holder.

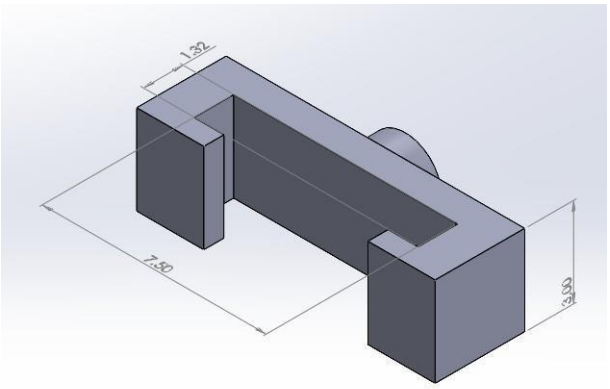


Figure 4.8 Camera Holder

4.3.8 Rotary Encoders

The rotary encoders are connected to the links through PVC clamps. They mesh with the concentric gears and provide the linear and angular coordinates as the links are moved. This Figure 4.10 depicts the image of the rotary encoder.

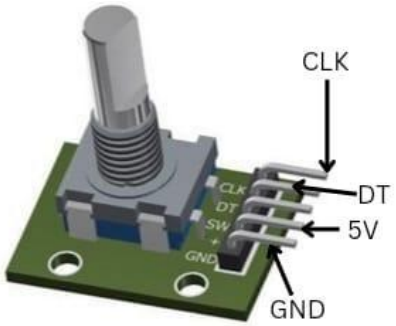


Figure 4.9 Rotary Encoders

4.3.9 Gears

The prototype consists of various gears of different sizes. The gears are meshed with the rotary encoders. This Figure 4.11 depicts the image of the gears.

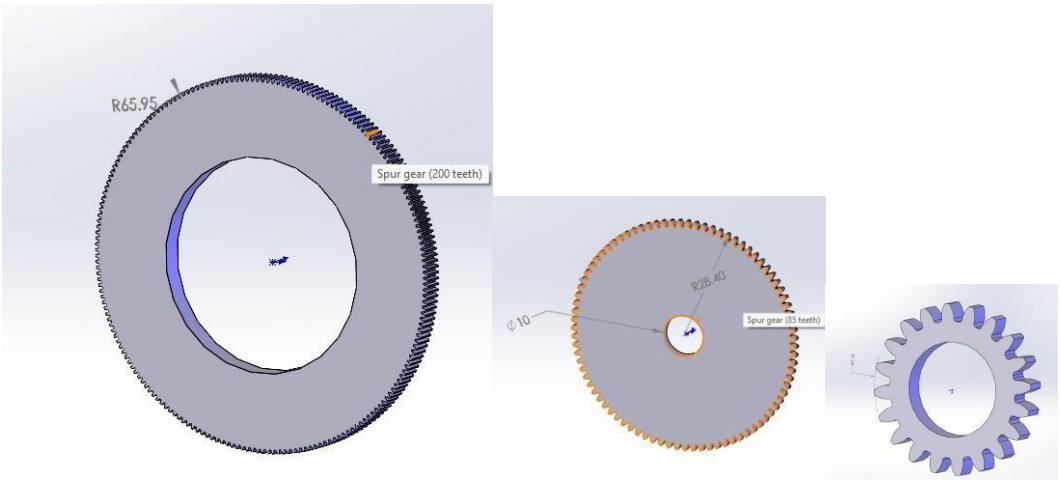


Figure 4.10 Gears

4.3.10 Final Design

Based on the above calculations, the final prototype specifications are:

- Structure Dimensions
  - Total Height: 167.5 cm
  - Maximum Radius: 136 cm
  - Base Diameter: 60 cm
  - Platform Diameter: 25 cm
- Linkage System
  - Number of Arms: 3
  - Link Length: 32 cm each
  - Joint Spacing: 4.2 cm (base), 4.7 cm (end)
  - Total DOF: 4
- Movement Range
  - Base Rotation: 360°
  - Arm 1 Angular Range: ±145°
  - Arm 2 Angular Range: ±145°
  - Arm 3 Angular Range: ±145°
- Working Envelope
  - Minimum Reach: 40 cm
  - Maximum Reach: 136 cm
  - Vertical Range: 31.5 cm to 167.5 cm
  - Effective Work Volume: >90 cm^3
- Mechanical Specifications
  - Material: Stainless Steel (linkages)
  - Base Material: Plywood
  - Bearing Type: Chrome Steel
  - Support Type: Roller with 4 wheels
- Position Tracking
  - Number of Encoders: 9
  - Encoder Resolution: 360 pulses/revolution
  - Position Accuracy: ±0.5°

All the above components, when assembled together, result in the following design:

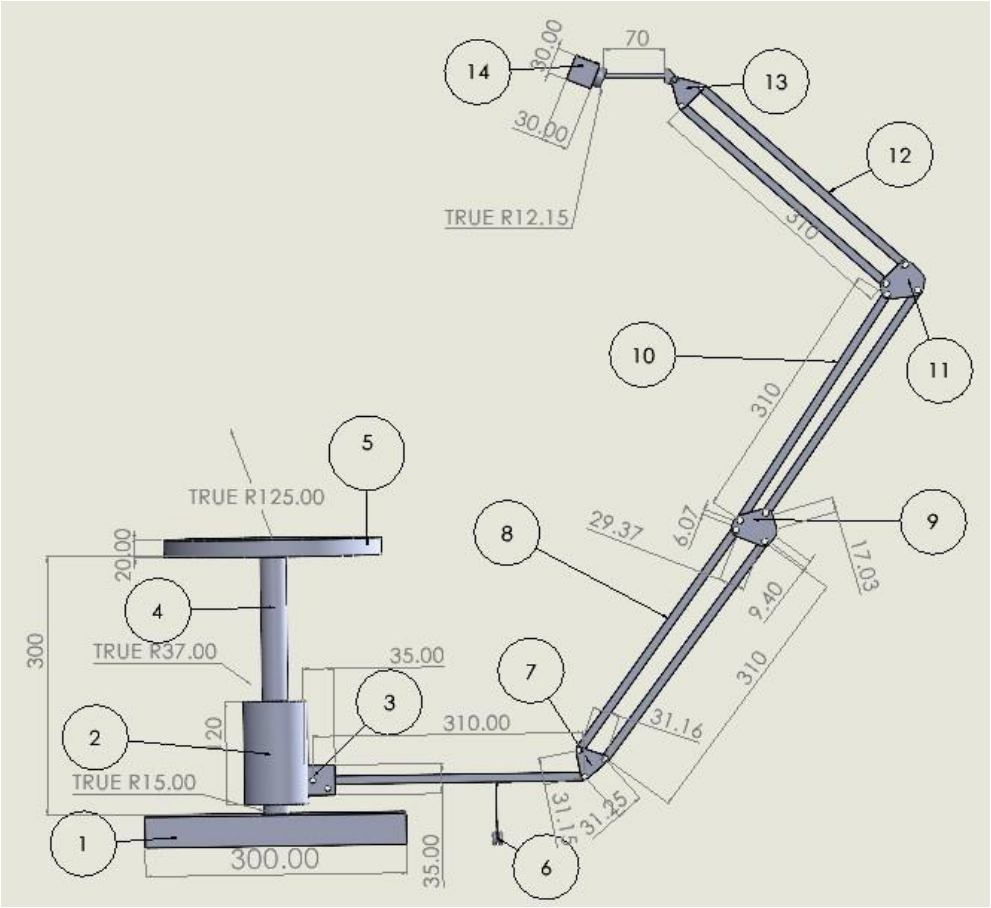


Figure 4.11 Completed of 4-DOF camera tracker design

4.4 MATERIAL SELECTION

The prototype consists of various component and each component is made of different materials. The Table 4.1 represents the material of each of component used in the prototype.

Table 4.1 Material Selection

COMPONENT	NOS	MATERIAL	SPECIFICATIONS	SOURCE	STATU
Base Plate	1	Plywood	30x20x3.60 Cm	Available At Home	Obtain
Bearing Casing	1	PVC Pipe	Diameter- 6.5 Cm	Available At Home	Obtain
Bearing	1	Chrome Steel	Od- 7cm Id- 3cm	Outsourced	Obtain
Circular Platform	1	Wood	25x2 Cm	Available At Home	Obtain
Shaft	1	PVC Pipe	30 Cm	Available At Home	Obtain
Linkages	7	Stainless Steel	32x1x1 Cm	Available At Home	Obtain
L Clamps	3	Stainless Steel	4 Cm	Available At Home	Obtain

Roller Support	1	PVC	2cm	Available At Home	Obtained
Camera Holder	1	Plastic	Retracted	Amazon	Obtained
Gears	13	Polymer	Diameter- 5.5 cm Diameter - 13 cm	Outsourced	Obtained
Rotary encoders	7			Outsourced	Obtained

4.5 FABRICATION OF PROTOTYPE

4.5.1 Assembly of the Apparatus with Components

The fabrication of a 4-DOF camera tracking system involves integrating key components to create a functional tool. One end of the shaft is connected to the base and the other end has a circular platform attached. The shaft passes through a bearing which is enclosed by a PVC casing. The apparatus consists of three links and joints. The horizontal arm at base is connected to the bearing casing with clamps. The entire linkage structure rotates along with the bearing casing. The Figure 4.11 represents the front view of the mechanical apparatus.

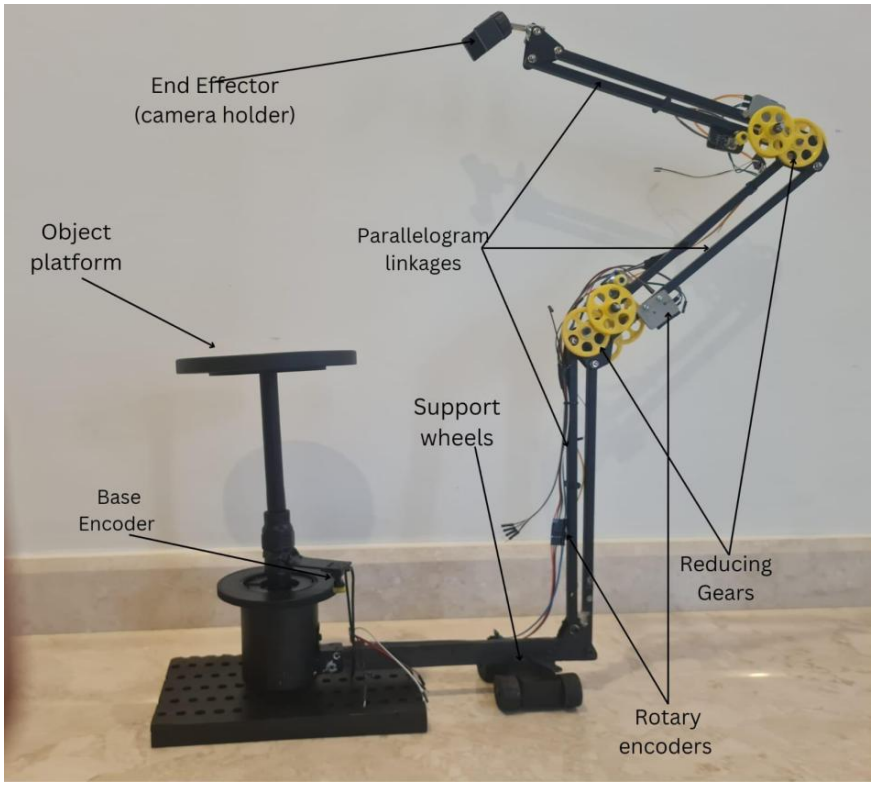


Figure 4.11 Front view of the apparatus

The rotary encoders are strategically placed in each of the six links and near the bearing casing. The stationery gear meshes with a concentric gear. The concentric gear further meshes with the gear of the rotary encoder placed in the link. The concentric gear is fixed by placing a shaft in the link. The similar setup is placed along six of the links.

A larger gear is fixed on the bearing casing and it rotates along with the bearing casing. A rotary encoder is fixed in the PVC clamp which is screwed to the shaft. The gear of the larger gear meshes with gear of the rotary encoder.

4.6 CIRCUIT DESIGN AND INTEGRATION

The electronic subsystem consists of seven digital rotary encoders strategically positioned across the mechanical linkage system to provide precise angular position data. These encoders are integrated with an Arduino Mega 2560 microcontroller board, which processes the signals and communicates with the host computer via USB serial connection.

4.6.1 Component Specification

The Table 4.2 represents the specifications of the microcontroller used to acquire data from encoders.

Table 4.2 Specification of Microcontroller

Platform	Arduino Mega 2560
Processor	ATmega2560
Operating Voltage	5V
Input Voltage	7-12V
Digital I/O Pins	54 (15 PWM outputs)
Clock Speed	16 MHz
Flash Memory	256 KB
SRAM	8 KB
EEPROM	4 KB

4.6.2 Pin Mapping Configuration

Base Rotation Encoder (E1):

The Table 4.4 represents the Configuration for Base Rotation Encoder

Table 4.4 Configuration for Base Rotation Encoder

CLK	D27
DT	D22
VCC	5V
GND	Common Ground

First Linkage Encoders (E2, E3):

The Table 4.5 represents the Configuration for First Linkage Encoder.

Table 4.5 Configuration for First Linkage Encoder

E2:

CLK	D28
DT	24

E3:

CLK	D29
DT	26

Second Linkage Encoders (E4, E5):

The Table 4.6 represents the Configuration for Second Linkage Encoder.

Table 4.6 Configuration for Second Linkage Encoder

E4:

CLK	D30
DT	48

E5:

CLK	D31
DT	46

Third Linkage Encoders (E6, E7):

The Table 4.7 represents the Configuration for Third Linkage Encoder.

Table 4.7 Configuration for Third Linkage Encoder

E6:

CLK	D32
DT	50

E7:

CLK	D33
DT	D1

4.6.2 Power Requirements

The Table 4.8 represents the Power Requirements of the microcontroller.

Table 4.8 Power Requirements

Total Current Draw	~450mA
Arduino Mega	100mA
Each Encoder	~35mA (×9)
Voltage Requirements	5V DC regulated

4.6.4 Power Distribution Network

The Table 4.9 represents the Power Distribution Network of the microcontroller.

Table 4.9 Power Distribution Network

Main Power Bus	5V from USB or external power supply
Ground Bus	Common ground plane for all components
Decoupling Capacitors	100µF electrolytic and 0.1µF ceramic per encoder
Power Distribution PCB	Custom designed for even power distribution

4.6.4. Signal Processing and Data Acquisation

Encoder Signal Processing

The Figure 4.13 represents the Code for Signal Processing of Encoder.

```
// Quadrature Decoding Algorithm
void processEncoderSignal(int encoderIndex) {
    int currentCLK = digitalRead(encoderPins[encoderIndex].clkPin);
    int currentDT = digitalRead(encoderPins[encoderIndex].dtPin);

    if (currentCLK != lastCLK[encoderIndex]) {
        if (currentDT != currentCLK) {
            encoderPositions[encoderIndex]++;
        } else {
            encoderPositions[encoderIndex]--;
        }
    }
    lastCLK[encoderIndex] = currentCLK;
}
```

Figure 4.13 Code for Signal Processing of Encoder

1. Function Signature:

void processEncoderSignal(int encoderIndex)

- **Purpose:** This function processes the signals from a quadrature encoder (identified by encoderIndex) to update its position.
- **Return Type:** void - This function doesn't return any value, but it modifies the encoder's position stored in a global or external variable.
- **Parameter:**
  1. encoderIndex: The index identifying which encoder's signal to process (useful if multiple encoders are in use).

2. Reading the Encoder Pins:

int currentCLK = digitalRead(encoderPins[encoderIndex].clkPin);

int currentDT = digitalRead(encoderPins[encoderIndex].dtPin);

- **Purpose:** The function reads two signals from the encoder:
  1. clkPin (Clock Pin): One of the two signals from the encoder.
  2. dtPin (Data Pin): The second signal from the encoder.
- **digitalRead:** This function reads the current digital state (HIGH or LOW) from the respective pin. These pins represent the two square wave signals of a quadrature encoder.

3. Checking for State Change:

```
if (currentCLK != lastCLK[encoderIndex]) {
```

- **Purpose:** The if condition checks whether the clock signal (currentCLK) has changed compared to the last time it was read (lastCLK). A change in this signal indicates movement.
- **lastCLK[encoderIndex]:** This array stores the last state of the clock pin for each encoder.

4. Determine Direction of Movement:

```
if (currentDT != currentCLK) {
  encoderPositions[encoderIndex]++;
} else {
  encoderPositions[encoderIndex]--;
}
```

- **Purpose:** Once a change in the clock signal is detected, the direction of movement is determined by comparing the data signal (currentDT) to the clock signal (currentCLK).

1. If currentDT != currentCLK, it means the encoder is moving in one direction (e.g., clockwise), so the position is incremented (++).
2. If currentDT == currentCLK, it means the encoder is moving in the opposite direction (e.g., counterclockwise), so the position is decremented (--).

5. Update Last Clock State:

lastCLK[encoderIndex] =

currentCLK;

- **Purpose:** After processing the signals, the current state of the clock pin (currentCLK) is saved as lastCLK. This ensures that the next time the function is called, it can compare the new state with this stored value to detect changes.

4.6.4 Data Acquisition Protocol

The Table 4.10 represents the protocols for data acquisition from encoders.

Table 4.10 Protocols for Data Acquisition

Sampling Rate	1000Hz per encoder
Resolution	0.25 degrees
Data Format	16-bit signed integer per encoder
USB Serial Communication	115200 baud rate

4.6.5 Cable Management

A 24 AWG shielded twisted pair cable is used with a maximum length of 50 cm. The Table 4.12 represents the requirements for cable management.

Table 4.11 Requirements for Cable Management

Cable Type	24 AWG shielded twisted pair
Cable Length	Maximum 50cm per encoder
Strain Relief	Integrated cable retention system
EMI Protection	Braided shield with proper grounding

4.6.6 System Calibration and Error Handling

Calibration Procedure:

- Initialize all encoders to zero position
- Perform full range of motion calibration
- Store offset values in EEPROM
- Validate position readings against known angles

Error Detection and Recovery:

The Figure 4.14 depicts the Code used for Error Detection of Encoder.

```
// Error Detection Algorithm
bool validateEncoderReading(int encoderIndex) {
    static long lastValidPosition[9] = {0};
    long currentPosition = encoderPositions[encoderIndex];

    // Check for unrealistic changes
    if (abs(currentPosition - lastValidPosition[encoderIndex]) > MAX_POSITION_CHANGE) {
        // Error detected - implement recovery
        encoderPositions[encoderIndex] = lastValidPosition[encoderIndex];
        return false;
    }

    lastValidPosition[encoderIndex] = currentPosition;
    return true;
}
```

Figure 4.14 Code for Error Detection of Encoder

1. Function Signature:

bool validateEncoderReading(int encoderIndex)

- **Purpose:** The function validateEncoderReading checks whether the current reading from an encoder (identified by encoderIndex) is valid.
- **Return Type:** bool - The function returns true if the reading is valid, and false if an error is detected.
- **Parameter:**
    - encoderIndex: The index of the encoder being validated (useful if there are multiple encoders being tracked).

2. Static Array Initialization:

```
static long lastValidPosition[7] = {0};
```

- **Static Variable:** The lastValidPosition array holds the last valid position of each encoder (up to 9 encoders). Declaring it static means that the array retains its values between function calls.
- **Initialization:** All values in this array are initially set to 0.

3. Current Position Retrieval:

```
long currentPosition = encoderPositions[encoderIndex];
```

- **Purpose:** This retrieves the current position from the encoderPositions array for the specified encoder (encoderIndex).

4. Error Detection Mechanism:

```
if (abs(currentPosition - lastValidPosition[encoderIndex]) >
MAX_POSITION_CHANGE)
```

- **Purpose:** The code checks if the change between the current position and the last valid position is larger than a predefined threshold

  (MAX_POSITION_CHANGE). If the change is too large, it is considered unrealistic or an error.
- **abs():** This function ensures the comparison is based on the absolute value of the change (so both positive and negative deviations are detected).

5. Error Handling:

```
encoderPositions[encoderIndex] = lastValidPosition[encoderIndex];
return false;
```

- **Purpose:** If an error is detected (i.e., the position change is too large), the code:

1. Restores the encoder position to the last known valid position.
2. Returns false to indicate an error has been detected.

6. Updating the Last Valid Position:

```
lastValidPosition[encoderIndex] =
currentPosition;
```

- **Purpose:** If no error is detected, the current position is considered valid, and the lastValidPosition for that encoder is updated to this value for future comparisons.

7. Return True (Successful Validation):

```
return true;
```

- **Purpose:** If no error was detected, the function returns true, signalling that the encoder reading is valid.

CHAPTER 5

TESTING AND EXPERIMENTATION

5.1 HARDWARE SETUP

The image depicts a quadrilateral arm designed with multiple linkages, each equipped with encoders for precise angle measurement. These encoders read the angular positions of their respective linkages and relay the data to an Arduino Mega microcontroller. The Arduino Mega processes the angle readings and converts them into a singular reading representing the orientation of an "imaginary arm" positioned between two linkages. This imaginary arm's angle is calculated relative to other similar imaginary arms within the quadrilateral system. The system also includes a rotating base to provide additional angular data, adding another dimension to the captured readings.

The encoders are securely connected through wiring to the Arduino Mega, which is placed on the base of the system. The system's design ensures robust and real-time data capture, transferring the calculated angles to a serial monitor every 50 milliseconds for further processing.

5.2 DATA PROCESSING AND COMPUTATION

Once the angle readings from the encoders are transferred to a computer via the serial monitor, a Python script processes the data. The script uses Denavit-Hartenberg (DH) parameters to convert the angular readings into spatial coordinates. These coordinates determine the position (x, y, z) and direction (θ, φ) of the camera. The Python script runs continuously, ensuring synchronization between the angle data and the video feed from a mobile camera.

The mobile camera, mounted on the quadrilateral arm, transmits live video to the computer using the DroidCam client. The Python script captures frames from the video feed every 10 seconds and associates each frame with its corresponding position and direction values. This information is compiled into a CSV file, which serves as a comprehensive dataset for subsequent reconstruction tasks.

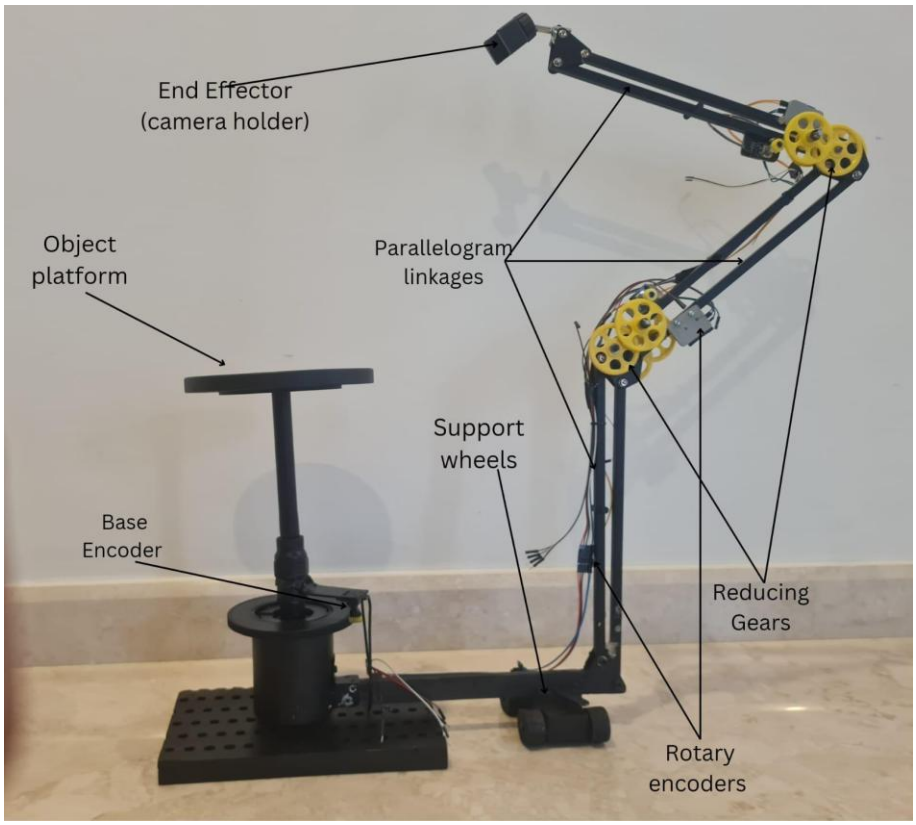


Figure 5.1 Fabricated Setup

5.3 RECONSTRUCTION AND OUTPUT

The generated CSV file is input into the NeRF (Neural Radiance Fields) Studio workspace, which is preloaded with the INGP (Instant Neural Graphics Primitives) model. NeRF utilizes the position and direction data from the CSV file, along with the captured video frames, to perform 3D reconstruction. This process results in a point cloud—a collection of points representing the spatial structure of the captured scene.

For practical applications, the point cloud can be further processed into a more usable format. An additional algorithm converts the point cloud into an STL (stereolithography) file, enabling 3D printing or further manipulation in CAD software.

CHAPTER 6

INTEGRATION WITH

AI

The AI model integrated with the camera tracking system is INGP (Instant Neural Graphics Primitives). INGP is a method that uses neural networks to quickly create and display detailed 3D objects and scenes. It works by efficiently storing and processing data, focusing on complex details, and training very fast. This makes it useful for tasks like 3D modeling, animations, and simulations that need accurate and smooth results.

6.1 DATA ACQUISITION

6.1.1 Methods of Data Collection in INGP

Data collection is the foundational step for the Instant Neural Graphics Primitives (INGP) framework, enabling it to process diverse types of input data for different graphics applications. INGP is inherently versatile, capable of handling data from multiple sources such as 2D images, volumetric grids, and signed distance functions. For tasks like scene reconstruction, INGP relies on Neural Radiance Fields (NeRF), which use 2D images captured from various viewpoints along with their camera poses. The Figure 6.1 depicts the acquisition of 2D images.

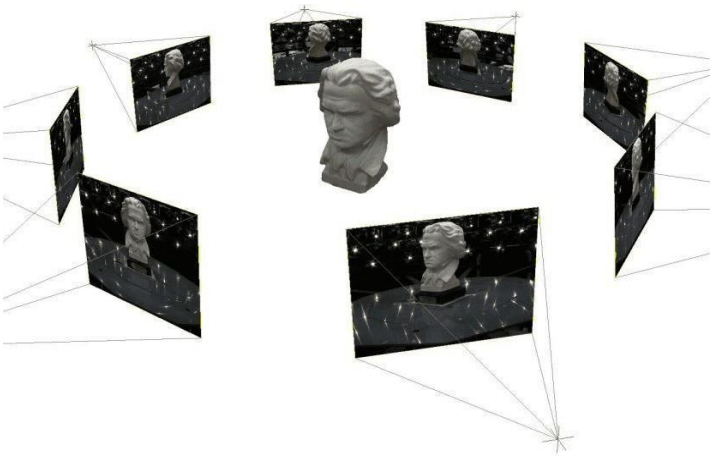


Figure 6.1 Acquisition of 2D images

These camera poses define the transformation between the image plane and the 3D coordinate space, allowing the system to reconstruct volumetric data. The richness of the acquired image dataset directly impacts the quality of the reconstruction; greater variation in angles and lighting conditions helps INGP capture fine details and complex textures.

For high-resolution imaging tasks, such as gigapixel image representation, INGP processes extremely large images with billions of pixels. Each pixel in such datasets corresponds to a unique coordinate and an associated RGB color value. This input format is critical for evaluating the system's ability to handle high-frequency details and accurately reconstruct fine textures. Unlike traditional methods that rely on dense grids for such tasks, INGP's multiresolution hash encoding enables efficient handling of these massive datasets, reducing memory usage while retaining high fidelity. The Figure 6.2 represents Ray Tracing for Scene Representation in INGP.

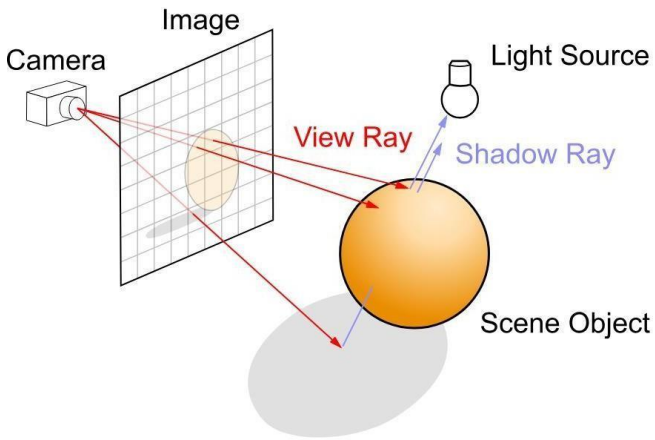


Figure 6.2 Ray Tracing for Scene Representation in INGP

For geometric modeling tasks, INGP works with Signed Distance Functions (SDFs). These datasets represent the shortest distance from a point in 3D space to the nearest surface, with negative, positive, or zero values indicating whether a point is inside, outside, or on the surface of the object. SDFs are particularly useful for representing smooth surfaces and sharp edges, which are essential in applications like CAD modeling and simulation. The adaptive nature of INGP's encoding allows it to allocate more resources to high- detail regions, such as sharp corners, without wasting computational effort on uniform regions of space.

6.1.2 Real-Time Data Processing for INGP

A distinguishing feature of INGP is its ability to process data in real-time, particularly in applications requiring dynamic interactions such as rendering. For example, in Neural Radiance Caching (NRC), the system dynamically acquires data by tracing light paths during the rendering process. Unlike precomputed datasets, which can be computationally expensive and inflexible, NRC utilizes real-time sampling techniques to collect data on-the-fly. This data includes spatial coordinates, light intensity values, and additional scene features such as surface normals and material properties. These features are critical for representing realistic lighting effects, including reflections and shadows.

INGP's multiresolution hash encoding plays a pivotal role in enabling real-time processing. By hierarchically encoding spatial data into compact hash tables, the framework minimizes the memory overhead while maintaining high accuracy. Adaptive sampling ensures that computational resources are focused on regions of interest, such as areas with intricate lighting effects or high curvature. This allows INGP to generate visually accurate outputs with minimal latency, making it highly suitable for interactive applications such as virtual reality, gaming, and live rendering in production environments.

6.2 PRE-PROCESSING

6.2.1 Multiresolution Hash Encoding in INGP

The multiresolution hash encoding method is a key pre-processing innovation in the INGP framework, enabling efficient representation of input data across varying spatial scales. In this method, the input domain is divided into hierarchical grids, each grid corresponding to a specific resolution level. Coarse grids capture large-scale structures, such as the overall shape of an object or the general layout of a scene. These grids store fewer points and are computationally inexpensive to process. On the other hand, fine grids focus on high-frequency details, such as sharp edges, fine textures, and small structural variations. By balancing the resolution levels, the encoding achieves a trade-off between efficiency and detail preservation.

At each resolution level, feature vectors are stored in hash tables, which serve as a compact representation of spatial data. Unlike traditional dense grids, which require significant memory and computation, the hash tables in INGP are lightweight and adaptive. For coarse levels, the mapping of grid points to hash table entries is direct, ensuring collision-free storage. For finer levels, where the number of grid points exceeds the hash table size, spatial hashing is employed. This approach uses pseudo-random hash functions to distribute grid points uniformly across the table, minimizing the chances of collisions. The hierarchical nature of this encoding enables INGP to handle large-scale scenes and high-detail objects with ease, making it suitable for a wide range of applications.

6.2.2 Collision Handling and Neural Optimization in INGP

Collision handling is a crucial aspect of INGP's multiresolution hash encoding, particularly at finer grid resolutions where multiple spatial

coordinates may map to the same hash table entry. Instead of traditional collision resolution methods like chaining or bucketing, which can increase memory overhead, INGP uses gradient-based optimization to resolve collisions implicitly. During training, the gradients of high-importance regions, such as visible surfaces or sharp boundaries, dominate the optimization process. This ensures that critical features are accurately represented in the hash table, even in the presence of collisions.

Furthermore, INGP employs linear interpolation to smooth transitions between adjacent grid points. This prevents blocky artifacts in the model's output, ensuring continuity across the spatial domain. Auxiliary inputs, such as view directions or material properties, are encoded separately using established methods like spherical harmonics or one-blob encoding. These features are then concatenated with the hash-encoded spatial data, providing a comprehensive representation for the neural network. This combination of multiresolution hashing, implicit collision resolution, and auxiliary feature encoding enables INGP to achieve rapid convergence and high accuracy in complex tasks.

6.3 3D MODEL GENERATION

6.3.1 NeRF within the INGP Framework

Neural Radiance Fields (NeRF) is one of the key applications of INGP, leveraging its capabilities to generate photorealistic 3D models. NeRF represents 3D scenes using a spatial density function, which encodes the occupancy of space, and a directional emission function, which captures the color and intensity of light emitted in different directions. The Figure 6.3 represents the Neural Rendering Pipeline for 3D Scene Reconstruction.

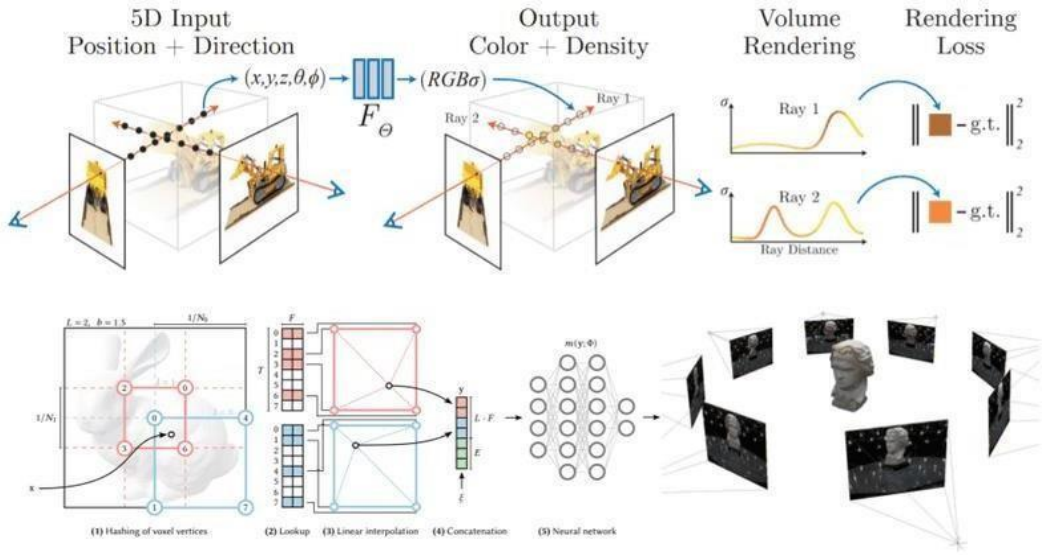


Figure 6.3 Neural Rendering Pipeline for 3D Scene Reconstruction

INGP enhances NeRF by encoding these functions hierarchically, using multiresolution hash encoding to balance global and local details. Coarse resolution levels in the hash table capture the overall shape of objects, while finer levels resolve intricate textures and lighting variations.

The training process involves optimizing the neural network to minimize the difference between rendered views and the input images. Differentiable ray marching is used to sample the density and emission functions along viewing rays, integrating these samples to compute the final pixel values. By using multiresolution hash encoding, INGP significantly accelerates this process, reducing training times from hours to seconds. This efficiency makes NeRF within INGP ideal for tasks such as virtual scene reconstruction, film production, and interactive visualization.

6.3.2 SDFs and Shape Modelling in INGP

Signed Distance Functions (SDFs) are another powerful application of INGP, particularly for precise geometric modelling. SDFs represent the shortest

distance from a point in 3D space to the nearest surface, enabling accurate modelling of sharp edges and smooth surfaces. INGP enhances the representation of SDFs through its hierarchical encoding system. Coarse resolution levels in the hash table capture the overall structure of the shape, while finer levels focus on high-curvature regions and small details.

The training process involves minimising the error between predicted and true distances for a set of sampled points. Adaptive sampling is used to prioritise regions with high geometric complexity, ensuring that critical features are accurately captured. Linear interpolation between adjacent grid points ensures smooth gradients, which are essential for applications like CAD, 3D printing, and real-time simulations. By combining SDFs with its efficient encoding system, INGP enables rapid and accurate geometric modelling, even for complex shapes.

CHAPTER 7

RESULTS AND CONCLUSIONS

7.1 COMPARISON OF RECONSTRUCTION METHODS

Table 7.1 summarises figures published by the authors of each method on their own benchmarks. [Editor's note, 2026: the original text said we tested these models under similar conditions. We did not; these are published values, and some do not match their sources, so the table should not be cited. Only Instant-NGP was run on our own captures (see Instant-NGP Performance below).]

Table 7.1 Published figures for reconstruction methods (not our measurements)

Model	Processing	Reconstruction Quality	Key Metrics
COLMAP (Traditional)	Hours to Days	High accuracy	-Not directly comparable with learning-based methods
Instant-NGP (INGP)	5-8 minutes	High quality	-PSNR: 33.09 -MIou: 0.629 (on ShapeNet)
NeRF (Original)	20-48 hours	High quality	-PSNR: 31.01 -SSIM: 0.947 -LPIPS: 0.081
Gaussian splatting (3DGS)	21-36 minutes	High quality	F-Score(T) 66.48% (on ShapeNet)

Mip NeRF 360	22-48 hours	Very high quality	-PSNR: 33.09 -SSIM: 0.961 -LPIPS: 0.043
Pienoxels	28 minutes	Good quality	PSNR: 23.06 (on Deep Blending dataset)

From the above table we can infer that,

- **Fast Processing (Minutes)**
  - Instant-NGP (5-8 minutes)
  - Gaussian Splatting (21-36 minutes)
  - Plenoxels (~28 minutes)
- **Moderate Processing (Hours)**
  - NeuS
  - Various MVSNet variants
  - Mesh-based methods
- **Slow Processing (Days)**
  - Traditional photogrammetry
  - Original NeRF
  - Mip-NeRF 360 (~48 hours)

7.2 CAMERA POSITION REGISTRATION PERFORMANCE

The mechanical apparatus demonstrated significant improvements in image registration compared to handheld capture methods:

7.2.1 Image Registration Statistics

- Handheld Video Capture (2-minute duration):
    - Trial 1: 52 images registered
    - Trial 2: 57 images registered
    - Trial 3: 66 images registered
    - Trial 4: 120 images registered
    - Average: 73.75 images registered
- Mechanical Apparatus Capture (2-minute duration):
    - Trial 1: 230 images registered
    - Trial 2: 252 images registered
    - Trial 3: 279 images registered
    - Trial 4: 277 images registered
    - Average: 259.5 images registered

This represents a 252% increase in successfully registered images using the mechanical apparatus compared to handheld capture.

7.2.2 Point Cloud Generation

Point cloud generation is the process of creating a collection of data points in 3D space, representing the surface geometry of an object or environment. The Figure 7.1 represents the Point Cloud of the model that is chosen for reconstruction.

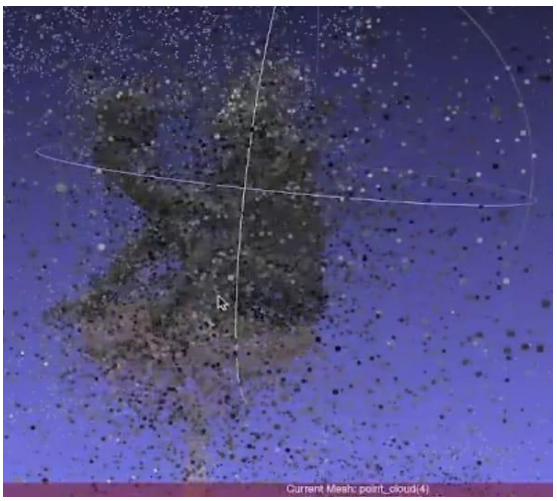


Figure 7.1 Point Cloud of the Model

- Optimal point cloud density was determined to be approximately 100,000 points per scene
- Higher point counts resulted in excessive noise
- Lower point counts led to insufficient detail in cropped models

7.3 CURRENT LIMITATIONS AND PROPOSED SOLUTIONS

7.3.1 Processing Pipeline Analysis

The most time-consuming steps in the current workflow are:

1. Structure from Motion (SfM) for camera pose estimation
2. Feature matching across multiple images
3. Dense reconstruction

7.3.2 Rationale for Rotary Encoder Integration

The integration of rotary encoders is expected to:

1. Eliminate the need for SfM camera pose estimation
2. Reduce processing time by providing direct angular measurements
3. Improve reconstruction accuracy through precise camera position tracking
4. Enable real-time camera position feedback

7.3.3 Instant-NGP Performance

Testing with Instant-NGP revealed:

- Initial rendering within 1.5 minutes
- Complete training in 7 minutes
- Optimal balance between processing speed and reconstruction quality

These results demonstrate the potential for significant improvement in 3D reconstruction efficiency through the combination of mechanical camera

position tracking and modern neural reconstruction methods. The planned integration of rotary encoders addresses the current bottleneck in the reconstruction pipeline while maintaining or improving reconstruction quality.

CHAPTER 8

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